Higher National Diploma in Engineering
04th Year, First Semester Examination - 2022
CE3223 – Water Supply & Waste Treatment

Instructions for Candidates:
Answer five questions only

No. of questions: 06
No. of pages: 03
Time: 03 hours

Q1).

a) It is necessary to know the rate of water to be supplied for a particular region because it varies from region to region. List four factors which affects the rate of demand. (4 marks)

b) List four objectives of community water supply system. (4 marks)

c) Find the population of a region after 3 and 4 decades by using following data. Use Arithmetical Increase method. (4 marks)

Year	Population
1935	10000
1945	10200
1955	12400

d) Briefly describe about "Hydrological cycle of water". (4 marks)

e) Comparative graphical method is a method to forecast population of a city. Briefly describe the method. (4 marks)

(Total 20 marks)

Q2).

a) Write two advantages and two disadvantages of ground water. (4 marks)

b) The main function of surface water intakes is to collect water and discharge for treatment. Write four essential requirements that should be considered to select a site for intake works. (4 marks)

c) State four physical characteristics of water. (4 marks)

d) Briefly describe two types of hardness in water. (4 marks)

e) What are the requirements that should be satisfied by an ideal indicator organism? (4 marks)

(Total 20 marks)

Q3).

a) What is meant by "Eutrophication". (2 marks)

b) State four effects of Eutrophication. (4 marks)

c) Briefly describe about "Coagulation" in water treatment process. (4 marks)

d) Mention four troubles available with sand filters. (4 marks)

e) What are the purposes of following units used in water treatment.

i. Coarse screens

ii. Raw water storage

iii. Pre chlorination

iv. Aeration

v. Plain sedimentation

vi. Flocculation (6 marks)

(Total 20 marks)

Q4).

a) There are several methods to distribute water to users. Gravity distribution method is one of the methods. Mention the advantages of that method. (4 marks)

b) State four requirements of water distribution system. (4 marks)

c) What are the factors that should be considered for selecting valves for a water distribution system? (4 marks)

d) State two advantages and two disadvantages of domestic water meter. (4 marks)

e) Briefly describe the reasons for corrosion of pipes. (4 marks)

(Total 20 marks)

Q5).

a) Calculate retention time of an aeration tank having 3.5 m x3.5 m x 2.5 m for 1000 m^3/day flow rate. (2 marks)

b) Compare and contrast the difference between trickling filters and activated sludge system used in wastewater treatment. (4 marks)

c) Calculate F/M ratio if the average daily flow is 20000 ml/day, BOD is 1000 mg/l, MLSS concentration is 200 mg/l and aeration basin dimensions are 2m x2m x 2.5m. (4 marks)

d) Calculate Sludge Volume Index in MLSS concentration is 2500 mg/l and settleable solid volume after 30 minutes is 125 ml in a activated sludge system. (2 marks)

e) Draw a cross section of a septic tank and mark important parts of it. (4 marks)

f) Briefly explain about "Oxygen Sag Curve". (4 marks)

(Total 20 marks)

Q6).

a) What are the factors that affect to generate wastes in a community? (4 marks)

b) State two advantages and two disadvantages of sanitary land filling method. (4 marks)

c) Briefly describe importance of National Environment Act. (4 marks)

d) List four health effects of noise pollution. (4 marks)

e) By assuming you are a construction company owner, State benefits of having ISO certification to the company. (4 marks)

(Total 20 marks)